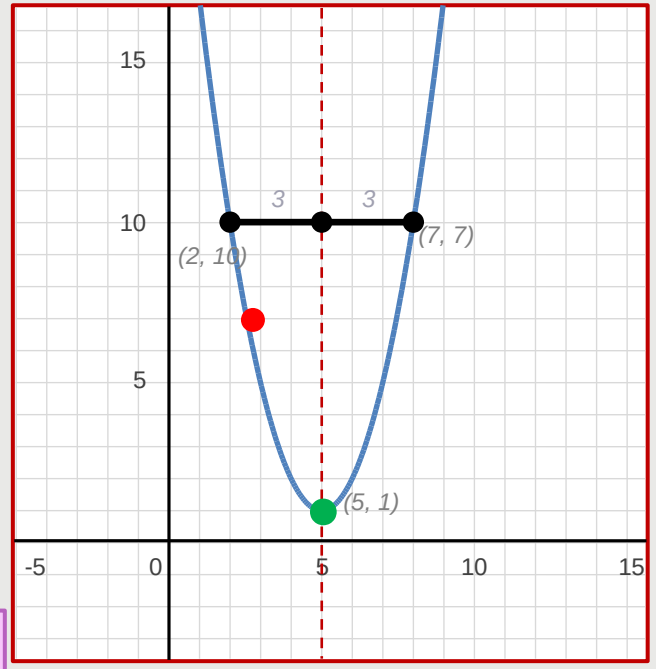
# Convex Function, Learning Rate =0.2, Start at X=3




| step | 0.2 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$f(x) = (x-5)^2 + 1$

$X = 5$

$f(x)$ Minimum value is **1.0**

***Reach Minimum after 11 Steps***

| x | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 5 | 4.24 | 3.56 | 2.96 | 2.44 | 2 | 1.64 | 1.36 | 1.16 | 1.04 | 1 |
| x-delta | 2.9 | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 |
| f(x-delta) | 5.41 | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 |
| x+delta | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 | 5.1 |
| f(x+delta) | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 | 1.01 |
| slope | -8 | -7.2 | -6.4 | -5.6 | -4.8 | -4 | -3.2 | -2.4 | -1.6 | -0.8 | 0 |
| next x | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 | stop |